STA314, Lecture 4

October 5, 2021

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The story so far

- Linear models are easy to fit (explicit formula).
- Performance can be similar to k-nn if regression function f is approximately linear.
- Typically do better than k-nn when there are many predictors.
- STA302: linear models are very interpretable and it is easy to do inference.
- Overall, 'basic' linear models as discussed in lectures do not have a very competitive prediction performance. However, they can provide good building blocks for more complex models that give better predictions - see coming lectures!

Multiple linear regression: some important statistics

The *total sum of squares* (poor man's training error)

$$TSS := \sum_{i=1}^{n}(y_i - \bar{y})^2, \quad \bar{y} := \frac{1}{n}\sum_{i=1}^{n} y_i.$$

The R^2 statistic (*aka proportion of variance explained*), for $\hat{b}$ linear regression estimator

$$R^2 := 1 - \frac{RSS(\widehat{b})}{TSS}, \quad RSS(b) = \sum_{i=1}^{n}(y_i - x_i^\top b)^2.$$

HW

Interpretation:

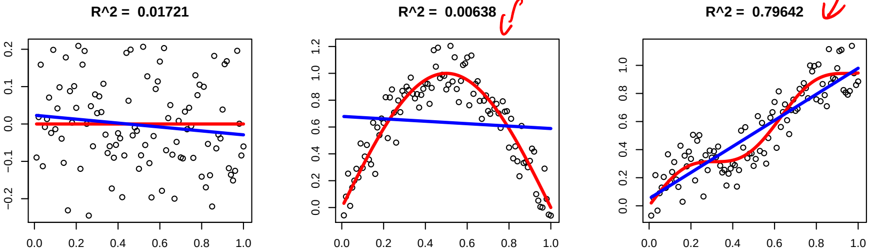
- $0 \leq R^2 \leq 1$. How do you prove that?
- $RSS(\hat{b})$ measures *training error* of prediction $\hat{f}(x) = \ldots$ $X^T\hat{b}$ (see lectures). TSS measures training error of prediction $\hat{f}(x) = \ldots$ (see lectures). $\bar{y}$
- If R^2 close to 1 that means $RSS(\widehat{b})$ 'a lot' smaller compared to TSS, i.e. linear model has much better prediction accuracy compared to 'poor man's prediction' $\bar{y}$ which ignores all predictors.
- Usually R^2 is taken as a measure of how much variation in y_i the linear model can explain. Some caution is needed when interpreting it...

$\Leftrightarrow$ RSS/TSS small

Interpreting R^2

does not mean

1. A small R^2 ... that the relationship between x and y is not linear.
2. A small R^2 ... that there is no relationship between predictor and outcome.
3. A large R^2 ... that the relationship is linear.



$y_i = 0 + \varepsilon_i$

Figure: Red curve: true regression function. Blue line: estimated linear regression function.

linear in x_i

$y_i = f(x_i) + \varepsilon_i$

Multiple linear regression in R: reading output

Running

`lm(Sales~TV)`

gives the output (assuming `Sales` and `TV` in workspace)

```
Call:
lm(formula = Sales ~ TV)

Residuals:
    Min      1Q  Median      3Q     Max
-8.3860 -1.9545 -0.1913  2.0671  7.2124

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 7.032594   0.457843   15.36   <2e-16 ***
TV          0.047537   0.002691   17.67   <2e-16 ***
---
Signif. codes:  0 *** 0.001 ** 0.01 * 0.05 . 0.1   1

Residual standard error: 3.259 on 198 degrees of freedom
Multiple R-squared:  0.6119,    Adjusted R-squared:  0.6099
F-statistic: 312.1 on 1 and 198 DF,  p-value: < 2.2e-16
```

Multiple linear regression in R: reading output

$f(TV) = b_1 + b_2 TV$

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 7.032594   0.457843   15.36   <2e-16 ***
TV          0.047537   0.002691   17.67   <2e-16 ***
---
Signif. codes:  0 *** 0.001 ** 0.01 * 0.05 . 0.1   1
```

- Coefficients: information about $\hat{b}$ in linear regression model.
- (Intercept): information about $\hat{b}_1$, the intercept
- TV: information the coefficient of predictor TV
- Estimate: value of $\hat{b}_1$, $\hat{b}_2$ etc

t value: value of t-test statistic. Pr($>|t|$): p-value of test for t-test. This is a test for $H_0 : b_j = 0$, i.e. predictor x_j *not informative in the presence of other predictors*.

Multiple linear regression in R: reading output

```
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Multiple R-squared:  0.6119,    Adjusted R-squared:  0.6099 
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```
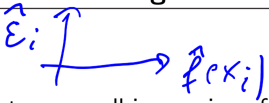
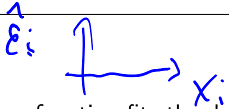

- Residuals: summary statistics about $\hat{\varepsilon}_i := y_i - x_i^\top \hat{b}$.
- Residual standard error (scaled square root of training error): $\sqrt{\frac{1}{n-d-1}\sum_{i=1}^{n} \hat{\varepsilon}_i^2}$
- Multiple R-squared: R^2 discussed earlier.

F-statistic and corresponding p value: for testing $H_0 : b_2 = \ldots = b_{d+1} = 0$, i.e. none of the predictors have linear effect on outcome (why this is called F test and what df means: see STA302).

$\iff$ $f(x) = b_1$

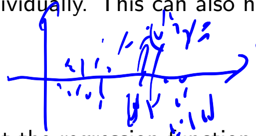
Multiple linear regression: residual plots





To get an overall impression of how well a regression function fits the data and if the linear model is correct, we often look at *residual plots*.

- If the predictor is one-dimensional, residuals $\widehat{\varepsilon}_i = Y_i - \widehat{f}(x_i)$ are plotted against x_i.
- If the predictor has more than one dimension, this is not feasible. In that case, residuals $\widehat{\varepsilon}_i$ are often plotted against predicted values $\widehat{f}(x_i)$.
- This will not always reveal deviations of the true model from linearity, so it is helpful to also plot against each predictor individually. This can also help with improving models.

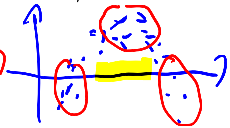


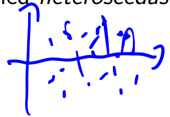
→ pattern in spread

Things to look for in a residual plot:

- *A pattern the mean of residuals* indicates that the regression function does not completely capture true relationship between predictor and outcome.
- *A pattern in the spread of residuals* indicates that the assumption of i.i.d. errors is not valid, this is sometimes called *heteroscedasticity*. Problematic for inference.

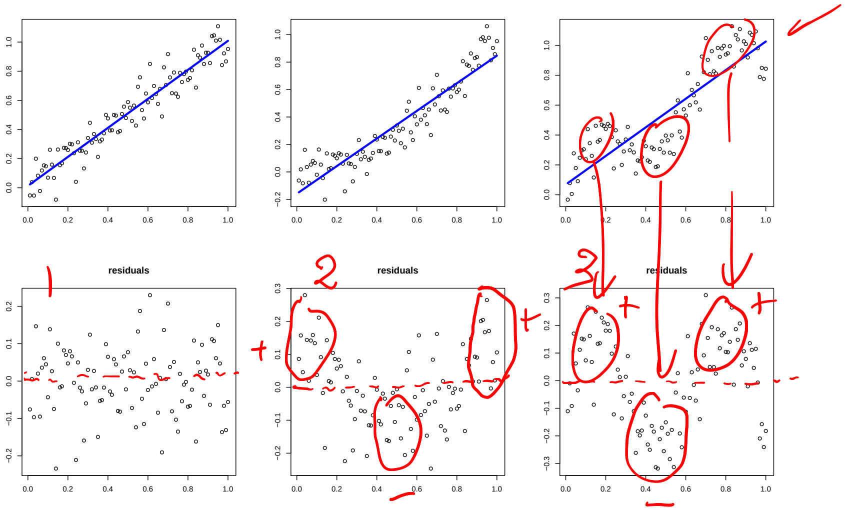
pattern





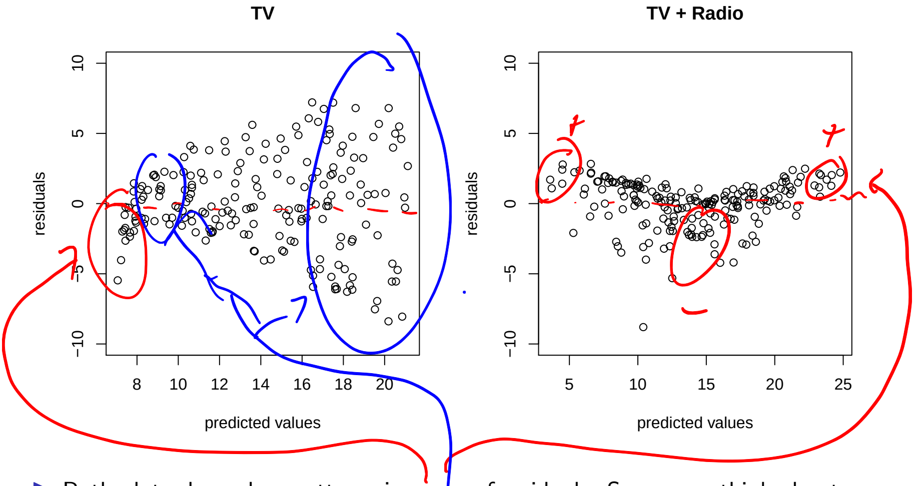
← no pattern in mean

Examples of residual plots: pattern in mean



- Upper row: data (dots) and estimated linear regression functions in blue.
- Lower row: residuals plotted against predictor values.
- Which plots show patterns in the mean of residuals? 2 + 3

Multiple linear regression: residual plots for Advertising data



- Both plots show clear patterns in mean of residuals. So we can think about further improving the model.
- Left plot shows clear pattern in spread of residuals, right plot shows some pattern in spread of residuals.

Multiple linear regression: interactions

So far we have only allowed each predictor to influence the response independently of the values of other predictors. This is not always realistic.

- Example: in the model $Sales = b_1 + b_2 \times TV + b_3 \times Radio + \varepsilon$, the effect of spending additional money on TV does not depend on how much we spent on Radio.

A simple way to model such effects while keeping the simple linear model structure is to allow for *interactions*. Including an interaction term between predictors $x_{i,1}, x_{i,2}$ means creating an additional predictor $x_{i,1}x_{i,2}$, i.e. a model of the form

$$f(x_i) = b_1 + b_2 x_{i,1} + b_3 x_{i,2} + b_4 x_{i,1} x_{i,2}$$

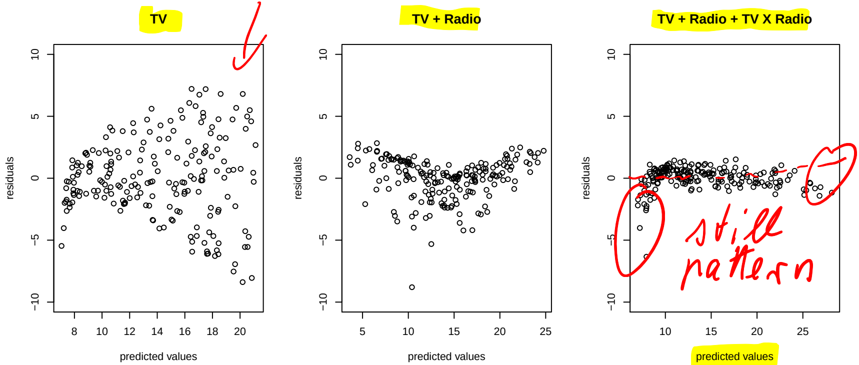
not linear in $x_{i,1}, x_{i,2}$
is linear in $b_1 \dots b_4$

Note: this changes the interpretation of b_2, b_3. Now increasing $x_{i,1}$ by one unit will increase $f(x_i)$ by $b_2 + b_4 x_{i,2}$ units!

- Example: for the Advertising data we could consider a regression function of the form

$$Sales = b_1 + b_2 \times TV + b_3 \times Radio + b_4 \times (TV \times Radio) + \varepsilon.$$

Example: Advertising data set with interaction terms



- Mean of squared residuals for 3 models: 10.51, 2.78, 0.87. **Note**: this is the training error. Is this surprising?
- Model with interaction still shows clear pattern in mean of residuals, so there is still something this model misses. Pattern in spread of residuals almost gone.

TV, Radio, TV ×Radio

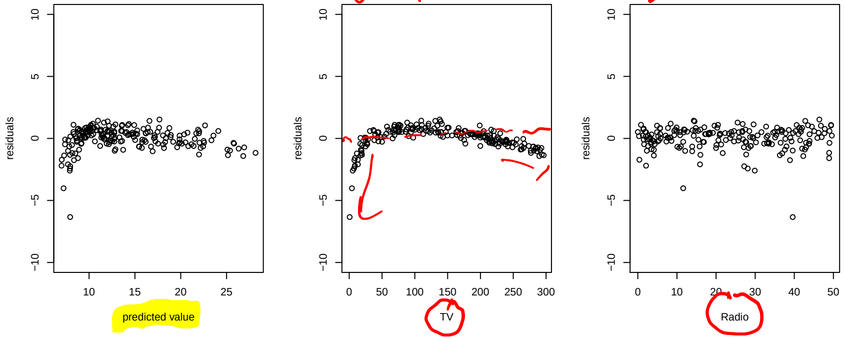
TV TV, Radio

no

Advertising data: a closer look at residuals

clear pattern in mean

no pattern in mean



- No pattern in residuals against `Radio`.
- Strong pattern in residuals against `TV`.
- Idea: allow for more flexible influence of `TV`.

Multiple linear regression: non-linear transformations of predictors

A popular way to allow non-linear effects of predictors on response is *polynomial regression of order k* where $f(x)$ is a polynomial of the form

d in x

$$f(x) = b_1 + b_2 x + b_3 x^2 + \dots + b_{k+1} x^k.$$

- this is still *linear in the coefficients* $b_1, \dots, b_{k+1}$.
- if we treat $x, x^2, \dots, x^k$ as new predictors this behaves like a multiple linear regression model with k predictors.

In general, any model of the form (recall: $x_i = (1, x_{i,1}, \dots, x_{i,d})^\top$)

$$f(x_i) = b_1 + b_2 g_1(x_i) + b_3 g_2(x_i) + \dots + b_{L+1} g_L(x_i)$$

for **given** functions $g_1, \dots, g_L$ and unknown coefficients $b_1, \dots, b_{L+1}$ inherits many properties of a linear model (can you write the linear model we considered so far in this form?). Another popular class of models is given by *basis expansions*, this will be discussed later in this course.

Examples of non-linear transformations of predictors

Which of the functions on this slide can be written as

$$f(x_i) = b_1 + b_2 g_1(x_i) + b_3 g_2(x_i) + \ldots + b_{d+1} g_d(x_i)$$

for known functions g_i

Example 1: $x_i = (1, x_{i,1}, x_{i,2})^\top$,

✓ $$f(x_i) = b_1 + b_2 x_{i,1} + b_3 x_{i,2} + b_4 x_{i,1} x_{i,2}$$

$g_1(x_i) = x_{i,1}$, $g_2(x_i) = x_{i,2}$

$g_3(x_i) = x_{i,1} x_{i,2}$

Example 2: $x_i = (1, x_{i,1}, x_{i,2})^\top$

✓ $$f(x_i) = b_1 + b_2 \sin(x_{i,1}) + b_3 \underbrace{\cos(1 + x_{i,2})} + b_4 x_{i,2}^3$$

$g_1(x_i) = \sin(x_{i,1})$ $g_2(x_i) = \cos(1 + x_{i,2})$ $g_3(x_i) = x_{i,2}^3$

Example 3: $x_i = (1, x_{i,1}, x_{i,2})^\top$

✗ $$f(x_i) = b_1 + \underbrace{\sin(b_2 + x_{i,1})} + b_3 x_{i,2}$$

not in form $b_2 g(x_i)$

R syntax for linear models

`X:Z`

is an interaction term between `X,Z`.

$X \cdot Z$

`X*Z`

means include `X,Z` and interaction `XZ`.

`I(X^k)`

means include $\mathtt{X}^k$ as predictor. Some other functions such as sin, cos, exp work directly but $I(\cdot)$ always works. Type `?formula` for details. See also function `poly` for polynomial regression.

If `D` is a data frame that contains a column with name `Y`

```
lm(Y ~ ., data = D)
```

will run a linear regression with Y as response and all other columns of `D` as predictors.

If `D` is a data frame that contains columns with names `Y`, `Z`

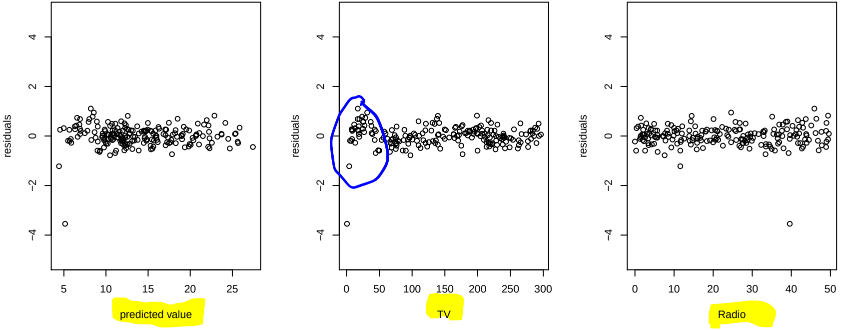
```
lm(Y ~ . - Z, data = D)
```

will run a linear regression with Y as response and all other columns of `D` **except Z** as predictors.

Non-linear transformation of predictors: Advertising data revisited

The residual plot of `Sales` against `TV` suggested that the effect of `TV` on `Sales` can not be modelled as a simple linear function. Try model

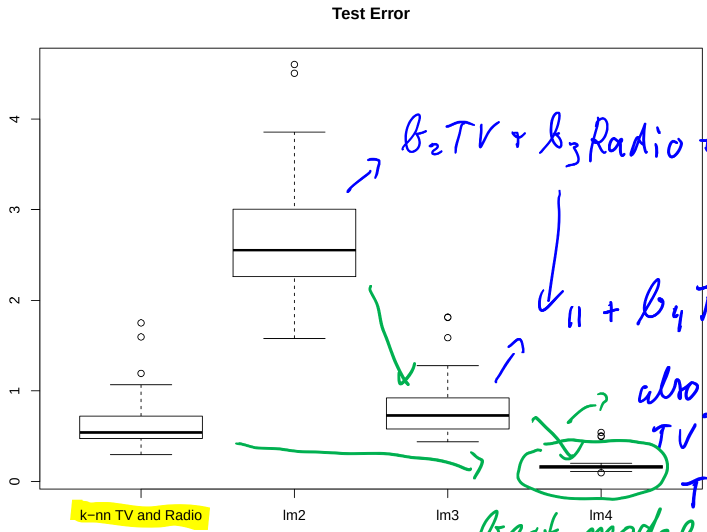
$$Sales = b_1 + b_2 \times Radio + b_3 \times (TV \times Radio) + \sum_{k=1}^{5} b_{3+k} \times TV^k + \varepsilon.$$



- Pattern in residuals almost disappeared.
- Mean of squared residuals decreased from 0.87 to 0.19, to be verified on test set.

careful: training error

Multiple linear regression vs K-nn: Advertising example



$\beta_2 TV + \beta_3 Radio + \beta_1$

$+ \beta_4 TV \times Ra$

also include TV^2, TV^3, TV^4, TV^5

best model

- lm2: additive form, lm3: included interaction, lm4: included interaction and non-linear transformation of `TV`. lm4 beats k-nn with `TV` and `Radio` by considerable margin

Motivation for next developments: Credit data set

The `Credit` data set contains the following information:

`Balance`
`Income`
`Limit`
`Rating`
`Cards`
`Age` (years)
`Education` `Gender` (Male/Female)
`Student` (Yes/No)
`Married` (Yes/No)
`Ethnicity` (Asian, Caucasian, African American)

Aim: predict `Balance` given information on the other variables.

`Gender`, `Married`, `Student`, `Ethnicity` are not numerical predictors, and it does not make sense to convert them to numerical predictors. Such variables are called *qualitative*.

Multiple linear regression: qualitative predictors

Motivation: so far we have considered numeric predictors. But sometimes predictors are qualitative (not numeric). Example: the `Credit` data set contains information data on student status (yes/no), ethnicity (African American, Asian, Caucasian) etc.

It does not make sense to treat student status or ethnicity as numeric variable. To include those variables in linear models, they are converted to *dummies*.

Example 1: `student` has two possible values. One possibility to convert this into dummy

$$x_i = \begin{cases} 1 & \textit{if} \quad \text{i'th person student} \\ 0 & \textit{if} \quad \text{i'th person not student} \end{cases}$$

Then

$$b_1 + b_2 x_i = \begin{cases} b_1 + b_2 & \textit{if} \quad \text{i'th person student} \\ b_1 & \textit{if} \quad \text{i'th person not student} \end{cases}$$

Interpretation:

- b_1 mean for non-students
- $b_1 + b_2$ mean for students
- b_2 is 'effect of being student'.

Multiple linear regression: qualitative predictors with $k > 2$ levels

Example 2: the credit data set contains a variable called `ethnicity` which takes values *African American*, *Asian*, *Caucasian*. Clearly no numerical ordering.

∨ Ex.: $k=3$

Idea: create $k-1$ dummies. For example

$$x_{i,1} = \begin{cases} 0 & \textit{if} \quad \text{i'th person not African American} \\ 1 & \textit{if} \quad \text{i'th person African American} \end{cases}$$

$$x_{i,2} = \begin{cases} 0 & \textit{if} \quad \text{i'th person not Asian} \\ 1 & \textit{if} \quad \text{i'th person Asian} \end{cases}$$

Then

$$b_1 + b_2 x_{i,1} + b_3 x_{i,3} = \begin{cases} b_1 & \textit{if} \quad \text{i'th person Caucasian} \\ b_1 + b_2 & \textit{if} \quad \text{i'th person African American} \\ b_1 + b_3 & \textit{if} \quad \text{i'th person Asian} \end{cases}$$

Running linear regression in **R** using the `lm` function will automatically convert qualitative predictors into dummies, *if qualitative predictors are not coded as numeric in advance.*

Variable selection

Motivation

- With Advertisement data, we determined the important predictors by 'trial and error'. This worked because there were only 3 predictors.
- The Credit data set has (after introducing dummies) 11 predictors. R output below indicates that not all are important.
- We will now look at systematic approaches to finding the important predictors.

is Education useful given everything else?

```
Call:
lm(formula = Balance ~ ., data = Credit)

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)        -479.20787   35.77394 -13.395  < 2e-16 ***
Income               -7.80310    0.23423 -33.314  < 2e-16 ***
Limit                 0.19091    0.03278   5.824 1.21e-08 ***
Rating                1.13653    0.49089   2.315   0.0211 *
Cards                17.72448    4.34103   4.083 5.40e-05 ***
Age                  -0.61391    0.29399  -2.088   0.0374 *
Education            -1.09886    1.59795  -0.688   0.4921
GenderFemale        -10.65325    9.91400  -1.075   0.2832
StudentYes          425.74736   16.72258  25.459  < 2e-16 ***
MarriedYes           -8.53390   10.36287  -0.824   0.4107
EthnicityAsian       16.80418   14.11906   1.190   0.2347
EthnicityCaucasian   10.10703   12.20992   0.828   0.4083
```

"large" p-values

↓

not useful predictors?

Selecting relevant predictors

If we have 'many' candidate predictors, how do we select 'the right ones'? Reasons for selecting fewer predictors:

Interpretation: if we can use fewer predictors to describe the response this helps to understand relationship between response and predictor and explain it to others. In the future, less data need to be collected and stored.

Improved prediction: in previous lectures we have seen that including many irrelevant predictors can lead to bad prediction performance.

Best subset selection

$\mathcal{M}_1$: model with smallest RSS among all with 1 predictor

Assume we have p candidate predictors. Best subset selection proceeds as follows

1. Denote by $\mathcal{M}_0$ a model with no predictors and just the intercept.
2. For $k = 1, 2, ..., p$
   2.1 For each set $\{j_1, ..., j_k\} \subset \{1, ..., p\}$ with exactly k elements compute RSS of linear model with predictors $x_{1,j_1}, ..., x_{1,j_k}$.
   2.2 Among models from step 2.1 select the one with smallest RSS. This is model $\mathcal{M}_k$.
3. From steps 1 and 2, we have *candidate models* $\mathcal{M}_0, \mathcal{M}_1, ..., \mathcal{M}_p$ with $0, 1, ..., p$ predictors. Select a model among those candidates, details later in this lecture.

- Motivation for step 2: models with same number of predictors have same flexibility.

How many models do we need to fit for this approach?

$\to 2^p$ $\to$ for each predictor: use or not use

Best subset selection

Assume we have p candidate predictors. Best subset selection proceeds as follows

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1. Denote by $\mathcal{M}_0$ a model with no predictors and just the intercept.
2. For $k = 1, 2, ..., p$
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- Motivation for step 2: models with same number of predictors have same flexibility.

How many models do we need to fit for this approach?

Potential issue: computational complexity when p is large.

- Doing cross-validation or the validation set approach in this setting right is tricky but very important. Comments on that later.

Forward stepwise selection

$1 + \frac{1}{2}p(p+1)$

1. Denote by $\mathcal{M}_0$ a model with no predictors and just the intercept.
2. For $k = 1, 2, \ldots, p$, starting with $\mathcal{M}_0$ do this iteratively
   2.1 Consider all possible $p + 1 - k$ predictors which are not in $\mathcal{M}_{k-1}$, this gives candidate models $\mathcal{M}_{k,1}, \ldots, \mathcal{M}_{k,p+1-k}$.
   2.2 Among models $\mathcal{M}_{k,1}, \ldots, \mathcal{M}_{k,p+1-k}$ select the one with smallest RSS, call it $\mathcal{M}_k$.
3. From steps 1 and 2, we have candidate models $\mathcal{M}_0, \mathcal{M}_1, \ldots, \mathcal{M}_p$ with $0, 1, \ldots, p$ predictors. Select a model among those candidates, details later in this lecture.

How many models do we need to fit for this approach?

1	$\mathcal{M}_0$
p	$\mathcal{M}_1$
$p-1$	$\mathcal{M}_2$
$\vdots$	
1	$\mathcal{M}_p$

$1 + \sum_{k=1}^{p} k =$

Ex.:

	A	B	C	A,B	A,C	B,C
RSS	10	7	6	3	4	5

$\mathcal{M}_0$ ✓ $\quad \mathcal{M}_1$: C $\quad \mathcal{M}_2$: A,C $\quad \mathcal{M}_3$: A,B,C

Forward stepwise selection

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   2.1 Consider all possible $p + 1 - k$ predictors which are not in $\mathcal{M}_{k-1}$, this gives candidate models $\mathcal{M}_{k,1}, ..., \mathcal{M}_{k,p+1-k}$.
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3. From steps 1 and 2, we have candidate models $\mathcal{M}_0, \mathcal{M}_1, ..., \mathcal{M}_p$ with $0, 1, ..., p$ predictors. Select a model among those candidates, details later in this lecture.

How many models do we need to fit for this approach?

- Less computation than best subset selection (difference can be substantial).
- Price for reducing computation: this is not guaranteed to not find the best model with given number of predictors.
- Doing cross-validation or the validation set approach in this setting right is tricky...

Backward stepwise selection

1. Denote by $\mathcal{M}_p$ a model which includes all predictors.
2. For $k = p-1, p-2, ..., 1, 0$, starting with $\mathcal{M}_p$ do this iteratively
   2.1 Consider all possible $k+1$ predictors which are in $\mathcal{M}_{k+1}$ and remove one at a time, this gives candidate models $\mathcal{M}_{k,1}, ..., \mathcal{M}_{k,k+1}$.
   2.2 Among models $\mathcal{M}_{k,1}, ..., \mathcal{M}_{k,k+1}$ select the one with smallest RSS, call it $\mathcal{M}_k$.
3. From steps 1 and 2, we have candidate models $\mathcal{M}_0, \mathcal{M}_1, ..., \mathcal{M}_p$ with $0, 1, ..., p$ predictors. Select a model among those candidates, details later in this lecture.

How many models do we need to fit for this approach? $\rightarrow$ HW

Backward stepwise selection

1. Denote by $\mathcal{M}_p$ a model which includes all predictors.

2. For $k = p-1, p-2, ..., 1, 0$, starting with $\mathcal{M}_p$ do this iteratively
   2.1 Consider all possible $k+1$ predictors which are in $\mathcal{M}_{k+1}$ and remove one at a time, this gives candidate models $\mathcal{M}_{k,1}, ..., \mathcal{M}_{k,k+1}$.
   2.2 Among models $\mathcal{M}_{k,1}, ..., \mathcal{M}_{k,k+1}$ select the one with smallest RSS, call it $\mathcal{M}_k$.

3. From steps 1 and 2, we have candidate models $\mathcal{M}_0, \mathcal{M}_1, ..., \mathcal{M}_p$ with $0, 1, ..., p$ predictors. Select a model among those candidates, details later in this lecture.

How many models do we need to fit for this approach?

- Similar amount of computation as forward stepwise.
- As forward stepwise, this is not guaranteed to not find the best model with given number of predictors.
- Doing cross-validation or the validation set approach in this setting right is tricky...